

SIT281 Cryptography - PBL B (Week 7)

1. Public Key Cryptography and Digital Signatures (Use RSA)
 - a. Generate two key pairs (Public Key/ Private Key) for A (Sender) and B (Receiver). Use SageMath or Wolframalpha for calculations
 - b. Encrypt a message (decimal number) from A to B and let B decrypt it
 - c. Sign a message from A to B and let B verify it
 - d. Combine Encryption/ Decryption and Sign/ Verify and show how this protocol ensures Confidentiality and Authentication and Non-repudiation
 - e. Using an example, show how RSA can be broken by factoring a composite number composed of two prime numbers
2. Symmetric Key Cryptography (**Note: If you make mistakes with 0s and 1s, do not try to correct them. Just keep on and complete the task**)
 - a. DES
 - i. Assume a 64 bit random bit stream and do the Initial Permutation.
 - ii. Run this output through one Round and obtain the Output. Assume a Round Key
 - iii. Show the cost of Brute Force Attack given that there is a computer that can run 10^{13} instructions per second
 - b. AES
 - i. Assume a 128 bit random number and run it through all four stages of AES and obtain the Output.
 - ii. Show the cost of Brute Force Attack given that there is a computer that can run 10^3 instructions per second
 - iii. Compare the Brute Force Attack costs of DES and AES
3. Galois Fields and LFSR. (**Note: If you make mistakes with 0s and 1s, do not try to correct them. Just keep on and complete the task**)
 - a. Let $A = (x^4 + x^3 + x^2 + x + 1)$ and $B = (x^3 + x^2 + 1)$. Find $A+B$, $A-B$, $A*B$ and A/B in $Z_2(x)$.
 - b. LFSR
 - i. Assume a 4 bit LFSR with any feedback function and a seed, and generate a random key stream
 - ii. What is the length of the generated key
 - c. Modern Stream Ciphers
 - i. Use the generated key to encrypt/ decrypt a message of the double size of the key and show the Encrypted output and the Decrypted output. (Hint: In modern stream ciphers, Encryption and Decryption are done by XORing the key and the input)